

Yiming Yang

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Department of Electrical and Computer Engineering, Michigan Technological University

EDUCATION

- Ph.D. in Electrical and Computer Engineering** Aug 2021 - Apr 2026
Michigan Technological University, Houghton, MI, USA
- Dissertation: Robust Autonomous Driving in Winter Weather: Perception and Integrated Control Under Adverse Conditions
- Master in Control Science and Engineering** Aug 2017 - Jul 2020
Chang'an University, Xi'an, China
- Thesis: Constrained Optimization and Distributed Model Predictive Control-Based Merging Strategies for Adjacent Connected Autonomous Vehicle Platoons
- Bachelor in Automation** Aug 2013 - Jul 2017
Chang'an University, Xi'an, China
- Graduated from the "Excellent Engineer" program

RESEARCH INTERESTS

- Autonomous driving & Robotics, Resilient Perception in Adverse Conditions
- Stochastic & Data-Driven Optimal Planning & Control

PUBLICATIONS

- 1. Geometric and Radiometric Enhancements of Vehicles in LiDAR Point Clouds with Snow Accumulation** Under Review.
Yiming Yang, Jeremy P. Bos,
Submitted to IEEE International Workshop on Metrology for Automotive (IEEE MetroAutomotive 2026)
- 2. Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection** Under Review.
Yiming Yang, Jeremy P. Bos,
Submitted to IEEE/RSJ International Conference on INTELLIGENT ROBOTS & SYSTEMS (IROS)
- 3. Tracking the Invisible: Self-Supervised Wheel Track Detection with Thermal-RGB Fusion** Under Review
Yiming Yang, Jeremy P. Bos,
Submitted to IEEE/RSJ International Conference on INTELLIGENT ROBOTS & SYSTEMS (IROS)
- 4. Aggressive Autonomous Control on Snow and Ice** Apr 2025
Yiming Yang, Jeremy P. Bos
SAE International Journal of Advances and Current Practices in Mobility (10.4271/2025-01-8040)
- 5. Constrained optimization and distributed model predictive control-based merging strategies for adjacent connected autonomous vehicle platoons** Nov 2019
Haigen Min, Yiming Yang, Yukun Fang, Pengpeng Sun, Xiangmo Zhao, 10.1109/ACCESS.2019.2952049

Open-Source Datasets & Tools

- ThermalTrack: Multi-modal self-supervised traversability estimation in whiteout conditions**
A dataset of 7,608 synchronized Thermal/VIS frames for resilient perception in low-visibility environments.
- WADS-3D: LiDAR-based object detection under heavy snow**
Features 2,050 labeled LiDAR scans and 30,104 3D bounding boxes for object detection in heavy snowfall.
- labelCloud-Enhanced: 3D bounding box labeling tool for point clouds**
High-efficiency 3D labeling with AI pre-labeling, spatial copy-paste, and enhanced viewport rotation.

RESEARCH EXPERIENCE

Ph.D. Candidate, Michigan Technological University Aug 2021 – Present
Proposal: Robust 3D Human Pose Estimation in Adverse Winter Conditions Under Review

Lead Author | Sponsor: Sony Research Award Program

- Proposed a Thermal-VIS fusion approach and the first multi-modal dataset for heavy snow to bridge critical gaps in 3D-HPE under low-visibility conditions.

SAE Autodrive Challenge II Jan 2023 – Present

Sponsor: General Motors & SAE International

- Architected the vehicle control interface and ROS-dSPACE communication pipeline via MicroAutoBox III.
- Developed safety-critical autonomy state machines with robust failure handling and safe-exit mechanisms.
- Led HIL/VIL validation and authored DVP&R documentation to track system-wide reliability.

Railroad Crossing Vehicle Warning (RCVW) System Aug 2021 – Dec 2022

Sponsor: Federal Railroad Administration (U.S. DOT)

- Led V2X implementation, including VBS/RBS/RSU configuration, crossing map creation, and field testing.
- Developed an HIL system to accurately replicate and validate real-world connected vehicle scenarios.
- Presented large-scale PTV VISSIM simulations of the RCVW application for the RLS and FRA websites.

UAS Data Integration for Transportation Infrastructure Nov 2021 – Dec 2022

Sponsor: Michigan Dept. of Transportation (MDOT)

- Curated datasets from UAV imagery (I-96 corridor) to train deep learning counting models.
- Derived Origin-Destination (OD) metrics and verified system accuracy against state benchmarks.

Automated Infrastructure Generation (OSM to VISUM/VISSIM) Aug 2021 – Dec 2022

Research Assistant | Dept. of Civil, Environmental, and Geospatial Engineering

- Developed automated pipelines to convert OpenStreetMap data into simulation-ready networks for PTV VISUM/VISSIM.
- Automated network connectivity, zone division, and spline-based connector generation, reducing modeling time from days to minutes.
- Conducted multi-scale OD analysis using Waymo (micro-level) and Weijio (macro-level) datasets.

Master's Student, Chang'an University Aug 2017 – Jul 2020

Autonomous Vehicle Control & Trajectory Tracking Jul 2018 – Jun 2020

Sponsor: National Key R&D Program / Shaanxi Province Key Projects

- Designed drive-by-wire algorithms for steering, propulsion, and braking using incremental PID.
- Deployed power supply system, embedded system for trajectory tracking algorithm and remote controller.
- Managed the gasoline-to-EV architecture transition and developed an MCU-based vehicle controller.

QLTD Intelligent Connected Expressway Testbase Apr 2019 – Sep 2020

Sponsor: QiLu Transportation Development Group

- Implemented V2X-based adaptive IDM and Forward Collision Warning systems using DSRC/C-V2X.
- Validated V2V functionalities through real-world highway platoon tests and remote-control system deployment.

TEACHING EXPERIENCE

Graduate Teaching Assistant Jan 2023 – Present

Course: Robotic Systems Enterprise (Michigan Tech)

- Graduate teaching assistant for teams: Controls & MathWorks, Safety & Testing
- Organized regular meetings and workshops, translating high-level goals into actionable tasks and sprints using SMART criteria, while providing ongoing technical support.
- **Controls Focus:** Guided deployment of steering/propulsion/brake/lighting control and trajectory-tracking algorithms using Model-Based Design (MBD) and Stateflow autonomy state machines.
- **Safety Focus:** Directed functional safety validation (SOTIF) and decoupled lateral/longitudinal controller testing for autonomous functions.

TALKS & PRESENTATIONS

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| 1. Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection | Feb 2025 |
| Robotics Colloquium presentation, Michigan Tech Robotics Initiative, 2026, Houghton, MI, USA | |
| 2. Aggressive Autonomous Control on Snow and Ice | Apr 2025 |
| Conference proceedings talk, World Congress Experience (WCX), 2025, Detroit, MI, USA | |
| 3. Deep Deterministic Policy Gradient based Cooperative Platoon Logitudinal Control Strategy | Jan 2021 |
| Conference presentation, Transportation Research Board (TRB), 2021, Virtual | |
| 4. Research on Automatic Emergency Line Change Control Strategy Based on MPC | Jun 2019 |
| Conference presentation, World Transport Convention (WTC), 2019, Beijing, China | |

PATENTS

- 1. Autonomous vehicle controllers** (NO: ZL 2019 2 0786357.5)
- 2. Autonomous vehicle platoon controller** (NO: ZL 2019 2 0784716.3)
- 3. Unmanned vehicle power system and unmanned vehicle** (NO: ZL 2019 2 1037928)

SELECTED HONORS & AWARDS

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| 1. 2nd Place, Concept design report & SRS - SAE/GM AutoDrive Challenge II (Year 4) | Jun 2025 |
| 2. 2nd Place, Highway Challenge - SAE/GM AutoDrive Challenge II (Year 2) | Jun 2023 |
| 3. 3rd Place, Overall Dynamics - SAE/GM AutoDrive Challenge II (Year 2) | Jun 2023 |
| 4. Excellent graduate of Chang'an University | Jun 2020 |
| 5. Excellent graduate of Chang'an University | Jun 2020 |
| 6. First-class Academic Scholarship 2017-2018 | Nov 2018 |
| 7. 2017-2018 University-level Outstanding Graduate Student | Nov 2018 |
| 8. Best Technology Award of 2018 i-VISTA Auto Driving Challenge | Aug 2018 |
| 9. First Prize of 2018 i-VISTA Auto Driving Challenge - Urban Traffic Scene Challenge | Aug 2018 |
| 10. Second Prize of 2018 China Robot Competition - General Service Robot Project | Aug 2018 |

SKILLS

Computational: Python, C/C++ , MATLAB/Simulink for algorithm development and data analysis
Control Systems: Real-time control design and implementation (dSPACE, ConfigurationDesk, ControlDesk)
Traffic Simulation: PTV Vissim, PTV Visum
Robotics: ROS, GAZEBO, dSPACE integration with ROS
Embedded Systems: Microcontrollers (STM32, Teensy, Arduino), V2X hardware configuration, HIL testing
Hardware: CAN bus analysis, CAD (SOLIDWORKS), and PCB design (Altium Designer)